

BUTT OF LEWIS LH, United Kingdom

WMO: 030250

Lat: 58.517N

Lon: 6.267W

Elev: 75

StdP: 14.66

Time Zone: 0.00 (EUW)

Period: 86-97

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	32.4	33.8	24.0	18.2	37.1	25.9	19.9	37.8	55.9	43.7	51.9	43.1	12.8	210	0.952

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.9	63.1	58.7	60.9	57.5	59.3	56.4	59.5	62.0	58.1	60.2	56.9	58.7	13.8	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.3	72.8	60.5	57.0	69.4	59.2	55.9	66.6	58.1	26.1	62.4	25.2	60.5	24.4	58.8	64.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
48.1	42.9	39.4	DB	28.9	71.3	1.6	2.1	27.8	72.8	26.8	74.1	25.9	75.2	24.8	76.8
			WB	27.5	63.1	1.8	0.9	26.2	63.8	25.1	64.3	24.1	64.8	22.8	65.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.3	41.6	40.9	41.8	43.8	48.4	51.6	55.0	54.7	52.2	48.7	45.3	42.6
	DBStd	6.04	3.58	3.56	3.61	3.21	3.79	3.36	2.80	2.48	2.94	3.50	3.42	3.76
	HDD50	1504	261	256	255	186	77	19	1	0	10	66	143	230
	HDD65	6480	726	676	720	635	516	402	310	320	383	507	590	695
	CDD50	502	0	0	0	2	26	67	156	145	77	25	3	1
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
	WSAvg	19.1	23.8	22.4	23.1	17.9	17.0	14.3	14.9	15.5	18.4	19.7	20.7	22.0
	PrecAvg	52.4	6.1	4.7	4.5	3.2	2.9	2.8	3.3	3.7	4.5	5.4	5.5	5.8
(15) Precipitation	PrecMax	67.8	11.2	10.9	9.2	4.8	6.3	5.0	6.2	8.0	7.5	8.7	9.2	10.9
	PrecMin	37.8	2.3	2.2	1.5	1.3	0.4	1.3	1.2	1.6	1.5	2.0	2.7	1.3
	PrecStd	6.7	2.2	2.5	1.8	0.8	1.4	1.1	1.1	1.5	1.5	1.9	1.4	2.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.2	50.4	51.8	55.5	65.0	65.0	66.9	65.4	62.2	57.4	54.0	52.8
		MCWB	49.8	48.4	49.7	50.1	58.1	58.3	61.2	60.3	58.5	54.1	52.8	50.7
	2%	DB	50.0	49.1	49.7	51.9	59.9	61.0	62.9	61.5	59.2	55.7	52.5	51.0
		MCWB	48.6	47.8	47.9	48.6	54.9	56.7	59.0	58.5	56.4	53.0	51.1	49.7
	5%	DB	48.7	47.7	48.4	50.1	55.9	58.1	60.9	59.5	57.6	54.6	51.3	49.9
		MCWB	47.3	46.6	46.8	47.6	51.8	55.2	57.8	56.8	55.1	52.2	49.5	48.8
	10%	DB	47.5	46.3	47.0	48.5	53.2	56.0	59.3	58.3	56.5	53.5	50.3	48.6
		MCWB	46.0	44.8	45.3	46.3	50.0	53.4	56.6	55.8	54.2	51.5	48.3	47.1
	0.4%	WB	50.5	49.3	50.1	51.8	58.5	59.5	61.8	61.1	59.8	55.4	53.4	51.3
		MCDB	50.9	49.7	50.8	54.4	64.1	63.1	65.4	63.9	61.4	56.4	53.8	52.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	48.9	48.2	48.4	49.5	55.3	57.2	59.6	59.2	57.3	53.9	51.6	50.1
		MCDB	49.6	48.8	49.5	50.9	59.2	60.6	62.1	60.9	58.4	54.9	52.2	50.8
	5%	WB	47.7	46.8	47.2	48.1	52.3	55.5	58.2	57.6	55.9	52.8	50.3	49.0
		MCDB	48.6	47.6	48.2	49.6	55.3	57.9	60.3	59.0	57.0	54.1	51.2	49.8
	10%	WB	46.3	45.1	45.7	46.6	50.4	54.0	57.0	56.4	54.8	51.8	48.9	47.6
		MCDB	47.3	46.1	46.8	48.2	52.3	55.8	58.8	57.9	56.1	53.4	50.2	48.5
	5% DB	MDBR	6.3	6.1	6.9	6.5	6.4	5.7	5.9	6.1	6.5	6.1	6.1	6.3
		MCDBR	7.1	6.5	7.9	8.5	11.7	9.4	8.9	8.1	8.1	6.7	6.3	6.4
		MCWBR	7.8	7.0	7.7	6.7	7.9	6.9	6.2	6.3	6.8	6.0	6.8	7.2
		MCDBR	7.1	6.1	7.6	7.1	10.0	9.0	8.4	7.9	7.4	6.1	6.3	6.6
(35) Mean Daily Temperature Range	5% WB	MCWBR	7.9	6.9	7.9	6.4	7.4	6.9	6.3	6.8	7.1	6.3	7.2	7.4
(40) Clear-Sky Solar Irradiance	taub		0.352	0.356	0.369	0.383	0.380	0.372	0.377	0.368	0.365	0.350	0.327	0.317
	taud		2.100	2.235	2.301	2.321	2.386	2.451	2.452	2.483	2.454	2.355	2.197	2.091
	Ebn at Noon		135	206	243	261	271	274	270	265	245	208	147	109
	Edh at Noon		17	24	30	34	34	33	32	29	26	21	15	13
(44) All-Sky Solar Radiation	RadAvg		99	283	634	1123	1499	1512	1401	1137	742	374	140	65
	RadStd		12	27	43	74	107	131	96	73	37	31	13	7

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (20 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page